

SAYLANI IT TRAINING

SQL Server — Class 1 Homework

Querying · Sorting · Filtering · Set Operators

Instructions

Write a SQL Server query to answer each question below using the BikeStores sample database.

Database: BikeStores Tables: production.products, production.brands, production.categories, sales.customers, sales.orders, sales.order_items, sales.staffs, sales.stores, sales.stocks

Topic Legend:

Querying

Sorting

Filtering

Set Operators

Q1 List all brand names from the brands table.

Querying

Your answer:

```
select brand_name from [production].[brands]
```

Q2 Show the product name and list price of all products, sorted from most expensive to cheapest.

Sorting

Your answer:

```
select
    product_name,
    list_price
from [production].[products]
order by list_price desc
```

Q3 Find all customers who live in the state of New York (NY).

Filtering

Your answer:

```
select * from [sales].[customers]
where state = 'NY';
```

Q4 Show only the top 5 most expensive products in the store.

Querying

Your answer:

```
select Top 5 * from [production].[products]
order by list_price desc
```

Q5

List all products with a price between \$200 and \$500, sorted by price ascending.

Filtering*Your answer:*

```
select * from [production].[products]
where list_price between 200 and 500 order by list_price asc
```

Q6

Find all customers whose last name starts with the letter "S".

Filtering*Your answer:*

```
select * from [sales].[customers]
where last_name like 'S%';
```

Q7

List all products that belong to category 6 (Mountain Bikes) OR category 7 (Road Bikes).

Filtering*Your answer:*

```
select * from [production].[products]
where category_id = '6' or category_id = '7'
```

Q8

Show all orders that have NOT been shipped yet (shipped_date is missing).

Filtering*Your answer:*

```
select * from [sales].[orders]
where shipped_date is NULL
```

Q9

Display product name, brand ID, and a computed column showing 15% off the list price, labeled "Sale Price".

Querying*Your answer:*

```
Select
    product_name,
    brand_id,
    list_price*0.15 as sale_price
from [production].[products]
```

Q10

Get a unique list of all cities where BikeStores customers live, sorted alphabetically.

Querying*Your answer:*

```
select distinct city from [sales].[customers]
order by city asc
```

Q11

List all staff members who are currently active (active = 1), showing their full name and email.

Filtering*Your answer:*

```
select
    first_name,
    last_name ,
    email
from [sales].[staffs]
where active = 1
```

Q12

Using UNION, get a combined list of all cities from both customers and stores (no duplicates), sorted A–Z.

Set Operators*Your answer:*

```
select city from [sales].[stores]
union
select city from [sales].[customers]
order by city asc
```

Q13

Using EXCEPT, find product IDs that exist in order_items but are NOT in the products table.

Set Operators*Your answer:*

```
select
    product_id
from [sales].[order_items]
except
select
    product_id
from [production].[products]
```

Q14

List all products from the year 2016 with a price greater than \$1,500, sorted by price descending.

Filtering*Your answer:*

```
select * from [production].[products]
where model_year = '2016' and list_price > 1500
order by list_price desc
```

Q15

BONUS: Using UNION ALL, show Trek (brand_id = 9) products from 2016 and Surly (brand_id = 8) products from 2017. Include product name, brand, year, and price. Order by year then price descending.

Bonus*Your answer:*

```
select
    product_id,
    brand_id,
    model_year,
    list_price
from [production].[products]
where brand_id='9' and model_year = '2016'
union all
select product_id, brand_id, model_year, list_price from [production].[products]
where brand_id='8' and model_year = '2017'
order by model_year desc , list_price desc
```